

Project Name : Gopher Industries Food Remedy API

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RE019 – Allergen Terminology and Precautionary Food- Labelling Research

1. Introduction

Food Remedy is designed to help users understand food products and make more informed choices. Because food allergies can lead to serious reactions, allergen information needs to be handled carefully and consistently. A system that incorrectly interprets an allergen statement could potentially give users a misleading impression about whether a product is suitable for them.

This research examines allergen terminology, mandatory allergen declarations and precautionary allergen labelling that are relevant to Food Remedy. It also considers how the application should differentiate between a confirmed allergen, possible cross-contact and situations where the available information is incomplete or uncertain.

For this assessment, the main regulatory context is Australia and New Zealand. The research is therefore based primarily on the Australia New Zealand Food Standards Code and guidance published by Food Standards Australia New Zealand (FSANZ).

Since 25 February 2024, food businesses have been required to comply with Plain English Allergen Labelling requirements. These rules were introduced to make allergen information clearer, more consistent and easier for consumers to identify (FSANZ 2025a).

2. Recognised Allergen Terminology

FSANZ requires certain foods and ingredients to be declared when they are present in food. Under the current requirements, prescribed allergen names include wheat, fish, crustacean, mollusc, egg, milk, lupin, peanut, soy or soybean, sesame and specific tree nuts such as almond, Brazil nut, cashew, hazelnut, macadamia, pecan, pistachio, pine nut and walnut.

Barley, oats and rye must also be declared when they contain gluten. Sulphites must be declared when they are added at a concentration of 10 mg/kg or more (FSANZ 2025a).

The current rules also require allergen declarations to use clear, standard terminology. Allergens appearing in an ingredient list must be identified using the required name and bold formatting. A separate allergen summary beginning with the word “Contains” is also required near the ingredient list (FSANZ 2025a).

This is particularly relevant to Food Remedy because ingredient descriptions may not always use the most obvious allergen name. For example, a product may contain whey or casein rather than simply listing the word milk. FSANZ specifically highlights the move toward declaring allergens using simple terms such as “milk” instead of relying only on technical ingredient terminology (FSANZ 2025c).

For Food Remedy, this means different ingredient terms should be mapped to a consistent recognised allergen category. For example:

Raw ingredient: whey powder
Normalised allergen: milk

3. Precautionary Allergen Labelling

Precautionary allergen labelling, often referred to as PAL, is different from a mandatory allergen declaration.

Common examples include:

- “May contain peanuts”
- “May contain traces of milk”
- “May be present: tree nuts”

These statements are used to indicate that an allergen may be unintentionally present because of cross-contact during manufacturing, handling, transport or other stages of production.

FSANZ states that PAL is voluntary and is not regulated in Australia and New Zealand in the same way as mandatory allergen declarations (FSANZ n.d.-a).

This distinction is important for Food Remedy. For example:

Contains peanuts

means the allergen is declared as present. However:

May contain peanuts

means peanut exposure is possible, but peanut is not necessarily an intentionally included ingredient.

Food Remedy should therefore avoid storing these two statements as if they mean exactly the same thing.

4. Categories of Allergen Information

A useful approach would be to classify allergen information according to the type and strength of the available evidence.

Information type	Example	Interpretation	Recommended Food Remedy treatment
Confirmed allergen declaration	“Contains milk”	Allergen is confirmed as present	Store as confirmed
Ingredient-based allergen evidence	“Whey powder”	Ingredient is derived from a recognised allergen	Map to milk and treat as confirmed evidence
Precautionary warning	“May contain peanuts”	Possible unintended presence	Store as precautionary/possible
Cross-contact statement	“May be present: sesame”	Potential manufacturing cross-contact	Display warning separately
Allergen-free claim	“Gluten free”	Product makes an absence-related claim	Store separately and interpret carefully
Missing information	No allergen information available	Safety cannot be determined	Mark as unknown
Conflicting or	Ingredients and	Evidence is uncertain	Mark as uncertain and require review

incomplete information	allergen fields disagree		
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This approach would prevent Food Remedy from treating all allergen-related information as a simple yes/no value.

5. Confirmed Allergens, Possible Cross-Contact and Unknown Information

5.1 Confirmed allergen

A confirmed allergen is supported by direct evidence that the substance is present in the food. This may come from:

- the ingredient list;
- an allergen summary statement;
- or a recognised ingredient that maps to an allergen.

FSANZ states that prescribed allergens must generally be declared when they are present as ingredients or components of food additives or processing aids (FSANZ n.d.-b). Food Remedy could therefore display a message such as “Contains milk” when sufficient evidence confirms the allergen.

5.2 Possible cross-contact

Possible cross-contact is different because the allergen has not necessarily been intentionally added. For example, “May contain peanuts” should be presented as a warning, but Food Remedy should preserve the fact that it is precautionary information rather than confirmed ingredient evidence.

The application could therefore use wording such as: “Possible peanut exposure – manufacturer states ‘may contain peanuts’.”

5.3 Unknown or missing information

One of the most important distinctions for Food Remedy is the difference between “no allergen detected” and “allergen information unavailable”. If the database does not contain sufficient allergen information, the system should not assume that the product is safe.

No allergen information available

should produce: “Allergen status could not be verified.” It should not automatically produce: “No allergens present.” Missing information represents uncertainty, not proof of absence.

6. Terminology Variations and Their Impact on Food Remedy

Food product data can describe the same allergen in several different ways. Milk-related information, for example, may appear as milk, milk powder, whey, casein or other dairy-derived ingredients. If Food Remedy only performs exact keyword matching, some relevant allergen evidence could be missed.

A normalisation layer would therefore be useful. The system could maintain a mapping between raw product terms and a recognised allergen category. For example:

whey → milk
casein → milk

The database should also preserve the original text instead of replacing it completely. This would provide traceability and allow developers to understand why an allergen classification was generated.

The same principle applies to precautionary wording. Statements such as “May contain peanuts”, “May be present: peanuts” and “Produced in a facility handling peanuts” should not automatically become the same type of evidence as “Contains peanuts”. The application should retain both the normalised allergen and the evidence type.

7. Situations Where Food Remedy Should Avoid a Definite Safety Conclusion

Food Remedy should avoid making a definite statement that a food is safe when there is insufficient information. This may occur where:

- the ingredient list is missing;
- allergen information is unavailable;
- the product record is incomplete;
- terminology cannot confidently be mapped;
- allergen fields contain conflicting information;
- only a precautionary warning is available;
- or product information may no longer reflect the current manufacturer label.

This is important because food formulations and labelling can change over time. Food Remedy should therefore assist users with interpreting information rather than presenting itself as a replacement for the current product label.

FSANZ guidance on precautionary labelling also reinforces the need for caution because PAL is voluntary and different products may use different approaches to assess cross-contact risk (FSANZ n.d.-a).

8. Recommended Backend and Database Representation

Rather than representing allergen information using only a boolean value such as true or false, Food Remedy could use a richer structure.

Confirmed allergen example

```
{
  "allergen": "milk",
  "status": "confirmed",
  "evidence_type": "ingredient",
  "raw_text": "whey powder"
}
```

Precautionary example

```
{
  "allergen": "peanut",
  "status": "possible",
  "evidence_type": "precautionary_statement",
  "raw_text": "may contain peanuts"
}
```

Unknown information example

```
{
  "allergen_status": "unknown",
  "reason": "insufficient allergen information"
}
```

This structure would allow the Backend and Database teams to distinguish between confirmed presence, potential cross-contact and missing information.

9. Recommendations for Backend and Database Teams

1. Use a consistent allergen vocabulary based on recognised FSANZ terminology.
2. Maintain mappings between ingredient terminology and recognised allergens, such as mapping whey and casein to milk.
3. Keep confirmed allergen information separate from precautionary statements.
4. Represent missing information explicitly as unknown rather than interpreting it as allergen-free.
5. Store the original label or ingredient wording alongside normalised allergen information.
6. Record how an allergen was identified, such as through an ingredient, a “Contains” statement, a precautionary statement or a derived mapping.
7. Avoid absolute user-facing safety claims when the available information is incomplete.
8. Treat precautionary statements as warnings for relevant users while preserving their distinction from confirmed allergen presence.
9. Use clear user-facing terminology: “Contains milk” for confirmed information, “May contain peanuts” for precautionary information, and “Allergen information could not be verified” for unknown information.
10. Periodically review allergen mappings and terminology as product data, manufacturing practices and regulatory guidance change.

It is also important to recognise that FSANZ provides specific exemptions for some highly processed ingredients derived from allergenic sources, such as fully refined soybean oil and certain wheat-derived glucose syrups. This means future allergen-mapping logic should not rely solely on the source ingredient without considering applicable exemptions (FSANZ n.d.-c).

10. Scope and Limitation

This research is intended to provide practical guidance for how Food Remedy could structure and interpret allergen information.

It does not certify individual products as safe for people with food allergies and should not be treated as medical advice. Food Remedy should continue to encourage users to check the latest manufacturer packaging and allergen declaration when making allergy-related decisions.

The recommendations are based primarily on the Australian and New Zealand regulatory context and may require adjustment if Food Remedy expands into other countries with different allergen-labelling requirements.

11. Conclusion

The research shows that allergen information cannot be accurately represented using a single yes/no classification.

Mandatory allergen declarations, precautionary statements and missing information each communicate different levels of certainty. Under the Australian and New Zealand framework, prescribed allergens must be declared using clear and standardised terminology, while precautionary statements such as “may contain” remain voluntary (FSANZ 2025a; FSANZ n.d.-a).

For Food Remedy, the most appropriate approach is to maintain distinct states such as confirmed, possible and unknown. The application should also normalise recognised ingredient terminology while retaining the original source text and evidence type.

These changes would give the Backend and Database teams a clearer structure for handling allergen information and would reduce the risk of presenting uncertain or incomplete data as a definite safety conclusion.

References

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